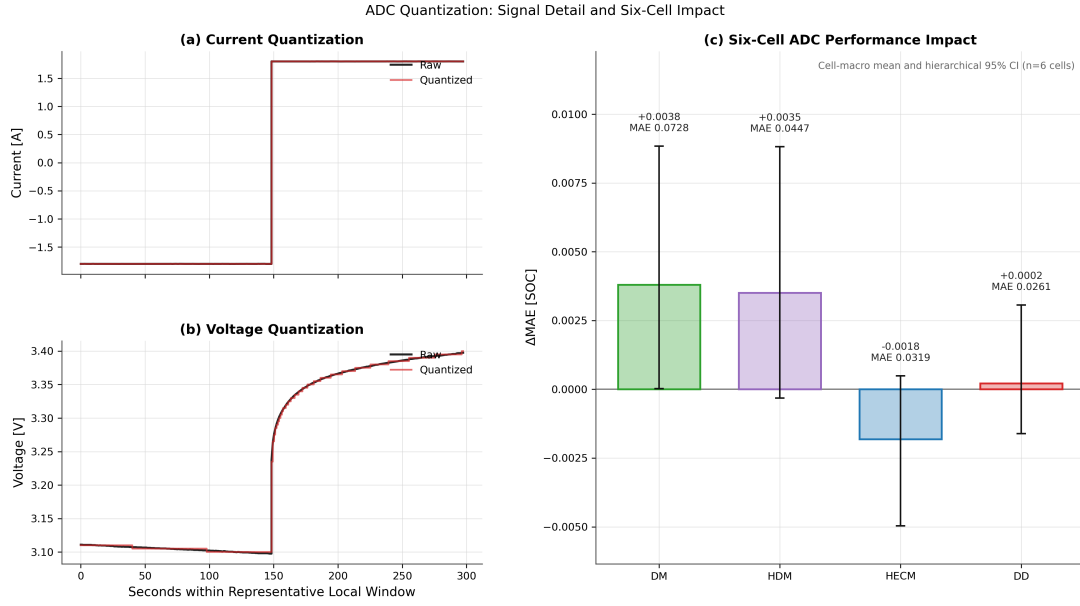




**Figure 6.14:** ADC quantization. Panels (a) and (b) compare representative raw and quantized current and voltage signals. Panel (c) reports six-cell  $\Delta\text{MAE}$  with hierarchical 95% confidence intervals and the resulting quantized-signal MAE.

#### 6.4.9 Diagnostic comparison with ground-truth SOH

The primary comparison supplies the same causally estimated LSTM-SOH trace to HDM, HECM, and DD. This is the health information available to the complete causal pipelines during online operation. The diagnostic comparison replaces it with dataset ground-truth SOH derived from capacity check-ups and interpolated between measured anchors. Its purpose is to determine whether SOC errors originate predominantly in the SOC branch or are inherited through the estimated SOH input.

Ground-truth SOH reduces baseline MAE by 0.0370 for HDM and 0.0166 for HECM, while DD MAE increases by 0.0049 on average. The pronounced HDM improvement follows from the direct use of SOH to scale effective capacity. HECM also benefits because SOH enters capacity scaling and lookup parametrization. DD uses SOH as one learned feature among several interacting inputs. Replacing that feature with its ground truth therefore does not impose a monotonic relation between SOH accuracy and SOC error.

The model-specific direction persists for most noise, sample-loss, and dropout cases. At 3% current-gain error, ground-truth SOH improves HDM and HECM and